

Q1. Void Function (No Return Value)

CODE

```
def welcome():  
    print("welcome to python programming")  
welcome()
```

OUTPUT

(no output)

Q2. Void Function (No Return Value)

CODE

```
def multiplication():  
    n=int(input())  
    for i in range(1,11):  
        print(n,"x",i,"=",n*i)  
multiplication()
```

INPUT

5

OUTPUT

(no output)

Q3. Void Function (No Return Value)

CODE

```
def even_numbers():  
    n=int(input())  
    for i in range(1,n+1):  
        if i%2==0:  
            print(i)  
even_numbers()
```

INPUT

40

OUTPUT

(no output)

Q4. Fruitful Function (Returns One Value)

CODE

```
def count_vowels(s):  
    count = 0  
    for ch in s.lower():  
        if ch in "aeiou":  
            count += 1  
    return count  
text = input()  
print(count_vowels(text))
```

INPUT

Programming

OUTPUT

(no output)

Q5. Fruitful Function (Returns One Value)

CODE

```
def largest_element(lst):  
    return max(lst)  
numbers = list(map(int, input().split()))  
print(largest_element(numbers))
```

INPUT 10 25 8 40 15

OUTPUT (no output)

Q6. Fruitful Function (Returns One Value)

CODE

```
def prime(n):  
    if n < 2:  
        return "Not Prime"  
    for i in range(2, n):  
        if n % i == 0:  
            return "Not Prime"  
    return "Prime"  
n = int(input())  
print(prime(n))
```

INPUT 7

OUTPUT (no output)

Q7. Function Returning Two or More Values

CODE

```
def student_result(m1, m2, m3):  
    total = m1 + m2 + m3  
    average = total / 3  
    if average ≥ 90:  
        grade = "A"  
    elif average ≥ 75:  
        grade = "B"  
    elif average ≥ 50:  
        grade = "C"  
    else:  
        grade = "F"  
    return total, average, grade  
m1 = int(input())  
m2 = int(input())  
m3 = int(input())  
total, average, grade = student_result(m1, m2, m3)  
print("Total =", total)  
print("Average =", average)  
print("Grade =", grade)
```

INPUT 85
90
95

OUTPUT (no output)

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Q8. Function Returning Two or More Values

CODE

```
def circle(radius):
    area = 3.14 * radius * radius
    circumference = 2 * 3.14 * radius
    return area, circumference
r = float(input())
a, c = circle(r)
print("Area =", a)
print("Circumference =", c)
```

INPUT

7

OUTPUT

(no output)

Q9. Function Returning Two or More Values

CODE

```
import math
def calculate(n):
    square = n ** 2
    cube = n ** 3
    sqroot = math.sqrt(n)
    return square, cube, sqroot
num = float(input())
s, c, r = calculate(num)
print("Square =", s)
print("Cube =", c)
print("Square Root =", r)
```

INPUT

19

OUTPUT

(no output)

Q10. Keyword Arguments

CODE

```
def student(name, age, department):
    print("Name:", name)
    print("Age:", age)
    print("Department:", department)
name = input()
age = int(input())
department = input()
student(name=name, age=age, department=department)
```

INPUT

Afreen
19
CSE

OUTPUT

(no output)

Q11. Keyword Arguments

CODE

```
def employee(empid, name, salary):  
    print("Emp ID:", empid)  
    print("Name:", name)  
    print("Salary:", salary)  
empid = int(input())  
name = input()  
salary = float(input())  
employee(name=name, salary=salary, empid=empid)
```

INPUT

```
101  
Afreem  
50000
```

OUTPUT

(no output)

Q12. Keyword Arguments

CODE

```
def book(title, author, price):  
    print("Title:", title)  
    print("Author:", author)  
    print("Price:", price)  
title = input()  
author = input()  
price = float(input())  
book(price=price, title=title, author=author)
```

INPUT

```
Python  
Guido  
450
```

OUTPUT

(no output)

Q13. Variable-Length Arguments (*args)

CODE

```
def add(*args):  
    print(sum(args))  
n = int(input())  
nums = []  
for i in range(n):  
    nums.append(int(input()))  
add(*nums)
```

INPUT

```
5  
10  
20  
30  
40  
50
```

OUTPUT

(no output)



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Q14. Variable-Length Arguments (*args)

CODE

```
def average(*args):  
    print(sum(args)/len(args))  
n = int(input())  
marks = []  
for i in range(n):  
    marks.append(float(input()))  
average(*marks)
```

INPUT

```
4  
80  
90  
70  
60
```

OUTPUT

(no output)

Q15. Variable-Length Arguments (*args)

CODE

```
def display(*args):  
    for i in args:  
        print(i)  
n = int(input())  
s = []  
for i in range(n):  
    s.append(input())  
display(*s)
```

INPUT

```
3  
Apple  
Banana  
Mango
```

OUTPUT

(no output)

Q16. Variable-Length Arguments (*args)

CODE

```
def largest(*args):  
    print(max(args))  
n = int(input())  
nums = []  
for i in range(n):  
    nums.append(float(input()))  
largest(*nums)
```

INPUT

```
5  
12  
45  
78  
23  
67
```

OUTPUT

(no output)

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Q17. Recursive Functions

CODE

```
def fib(n):
    if n <= 1:
        return n
    return fib(n-1) + fib(n-2)
n = int(input())
for i in range(n):
    print(fib(i), end=" ")
```

INPUT

7

OUTPUT

(no output)

Q18. Recursive Functions

CODE

```
def digit_sum(n):
    if n == 0:
        return 0
    return n % 10 + digit_sum(n // 10)
n = int(input())
print(digit_sum(n))
```

INPUT

12345

OUTPUT

(no output)

Q19. Recursive Functions

CODE

```
def reverse(s):
    if len(s) == 0:
        return s
    return reverse(s[1:]) + s[0]
text = input()
print(reverse(text))
```

INPUT

hello

OUTPUT

(no output)

Q20. Recursive Functions

CODE

```
def power(x, n):
    if n == 0:
        return 1
    return x * power(x, n-1)
x = int(input())
n = int(input())
print(power(x, n))
```

INPUT

2
5

OUTPUT

(no output)